

DEPARTMENT OF PHYSICS
BHABHA UNIVERSITY, BHOPAL
M. Sc. (Physics)

PROGRAMME SPECIFIC OUTCOME (PSO):

The academic programs of the department have been designed to cater the scientific and technological manpower requirements at the national and international arena. The study curriculums for these courses have been designed in tune with the changing scenarios. It also incorporates latest technological developments, societal relevance, global interface and self-sustaining learning. In order to impart teaching where the students would achieve depth of knowledge, we supplement the class room teaching of the regular syllabi with tutorials pertaining to each theory paper. These tutorials consist of exploration of brainstorming ideas and problems associated with specific topics of relevance. Similarly, an experimental skill for the maintenance of sophisticated equipment is developed through designing new experiments and devices as a part of laboratory courses. The concept of minor / major projects has been introduced for students to adventure into preliminary research field both in theory and experiment. In the project, the student has to explore new developments from books and journals collecting literature / data, carry out some innovative work and write a dissertation on its basis. As co-curricular activity regular seminars, invited talks, visits of eminent dignitaries etc are arranged. The students are also encouraged through proper counseling to undergo summer trainings in the globally reputed research laboratories and multinational industries. The department can take a genuine pride on its students who are placed all over the country in many scientific organizations like DRDO, NDA, BARC, CSIR laboratories; academic institutes like several universities/colleges and celebrated MNCs. The skilled manpower developed by the department is also scattered across many countries all over the world.

The Master of Science in Physics program is evenly distributed across four semesters. The first three semesters consist of four theory and two practical papers. In the fourth semester one theory paper is replaced by project work. Each paper shall be of 100 Marks out of which 20 marks are allotted to internal assessment and 80 marks for University examination. Minimum pass marks are 10 for the internal assessment and 32 for the University examination. In

addition, seminar and assignment will be 100 marks each and passing marks are 50 in each.

The semester-wise pattern is given below-

SEMESTER I:

Paper/Units		Internal Marks	Exam. Marks
MPH -511	MATHEMATICAL METHODS IN PHYSICS	20	80
Unit – I	Complex Variables		
Unit – II	Differential Equations		
Unit – III	Integral Transforms		
Unit – IV	Computational Methods in Physics		
Unit- V	Tensors		
MPH -512	QUANTUM MECHANICS	20	80
Unit – I	Basic Concepts		
Unit - II	Operators and Matrix Theory		
Unit - III	Approximation Methods		
Unit - IV	Stationary Perturbation Theory		
Unit – V	Time Dependent Perturbation Theory		
MPH -513	CLASSICAL MECHANICS	20	80
Unit – I	Lagrangian Mechanics		
Unit – II	Conservation and Variational Principle		
Unit - III	Scattering and Collision		
Unit - IV	Rigid Body Motion		
Unit- V	Canonical Equation		
MPH -514	ANALOG and DIGITAL ELECTRONICS	20	80
Unit - I	Differential and Operational Amplifier		
Unit - II	Practical Operational Amplifier		
Unit - III	Op-amp Oscillator		
Unit - IV	Gates, Flip flop and Counters		
Unit V	Microcomputer		
LABORATORY COURSES			
MPH -515	Physics Lab.-I (Based on MPH 512 & MPH 513)	50	
MPH -516	Physics Lab.-II (Based on MPH 514)	50	
Total		500	

SEMESTER II:

PAPER/UNITS		Internal Marks	Exam. Marks
MPH- 521	ELECTROMAGNETISM	20	80
Unit – I	Polarization		
Unit – II	Boundary Problems		
Unit – III	Magnetization		
Unit – IV	Electromagnetic Field		
Unit- V	Relativistic Mechanics		
MPH- 522	CONDENSED MATTER PHYSICS	20	80
Unit – I	Crystallography		
Unit – II	Mechanical Properties		
Unit – III	Lattice Dynamics		
Unit - IV	Electronic and Thermal Properties		
Unit – V	3D lattice vibrations		
MPH- 523	STATISTICAL MECHANICS	20	80
Unit – I	Basic Concepts		
Unit – II	M-B distribution		
Unit - III	B-E Statistics		
Unit - IV	Phase Transitions - I		
Unit- V	Phase Transitions-II		
MPH- 524	ADVANCED QUANTUM MECHANICS	20	80
Unit - I	Many Electron Problem		
Unit - II	Scattering		
Unit - III	Identical Particles		
Unit - IV	Radiation		
Unit V	Electromagnetic Potentials		
LABORATORY COURSES			
MPH- 525	Electronic Instrumentation Practical (Based on MPH 521-522)	50	
MPH- 521	General Physics II Practical(Based on MPH 523-24)	50	
Total		500	

SEMESTER III:

PAPER / UNITS		Internal Marks	Exam. Marks
MPH -531	NUCLEAR PHYSICS	20	80
Unit – I	Binding Energy		
Unit – II	Nuclear Forces		
Unit – III	Nuclear Reactions		
Unit – IV	Detectors		
Unit- V	Elementary Particles		
MPH -532	ATOMIC AND MOLECULAR PHYSICS	20	80
Unit – I	Basic Topics		
Unit - II	Zeeman Effect		
Unit - III	Types of Molecules		
Unit - IV	Diatomic Molecules		
Unit – V	Raman Effect		
MPH-533	ADVANCED CONDENSED MATTER PHYSICS	20	80
Unit – I	Superconductivity		
Unit – II	Magnetism		
Unit - III	Dielectric and Ferroelectric		
Unit - IV	Semiconductors I		
Unit- V	Semiconductor II		
MPH -534	DIGITAL SYSTEM AND MICROPROCESSOR	20	80
Unit - I	D/A, A/D converter		
Unit - II	Memory and Microprocessor		
Unit - III	Microprocessor Programming		
Unit - IV	Microprocessor Interfacing		
Unit V	Data Communication		
OR			
MPH- 534	SEMICONDUCTOR DEVICES	20	80
Unit - I	Semiconductors		
Unit - II	Diode and Transistors		
Unit - III	Unipolar Devices		
Unit - IV	Rectifiers and Converters		
Unit V	Machines and Motors		
LABORATORY COURSES			
MPH -535	Microprocessor Programming Practical	50	
MPH- 536	Computer Applications Practical	50	
Total		500	

SEMESTER IV:

PAPER/UNITS		Int. Marks	Exam. Marks
MPH- 541 ELECTRONIC DEVICES and INSTRUMENTATION		20	80
Unit – I	Semiconductor Devices		
Unit – II	Photonic Devices		
Unit – III	Measurement Properties		
Unit – IV	Instrumentation Design		
Unit- V	Vacuum Techniques		
MPH -542 FIBER OPTICS AND LASER PHYSICS		20	80
Unit – I	Fiber Optics- I		
Unit - II	Fiber Optics -II		
Unit - III	Laser Theory -I		
Unit - IV	Laser Theory -II		
Unit – V	Types of Lasers		
OR			
MPH -543 MICROWAVE AND OPTICS COMMUNICATION		20	80
Unit – I	Microwave Generators		
Unit – II	Oscillators		
Unit - III	Microwave Components		
Unit - IV	Fiber Optics Communication I		
Unit- V	Fiber Optics Communication II		
MPH -544 MATERIALS SCIENCE		20	80
Unit - I	Phase Transitions		
Unit - II	Kinetics and Diffusion		
Unit - III	Nanomaterials		
Unit - IV	X-Ray Diffraction		
Unit V	Microscopy Techniques		
LABORATORY COURSES			
MPH- 545 Materials Science Practical & Laser Optics Practical		50	
MPH -546 Project Work		100	
Total		500	

COURSE SPECIFIC OUTCOME (CSO) Ist SEMESTER

The first semester course has been designed to impart knowledge in core Physics subjects. The conventional papers like Mathematical Physics, Quantum Mechanics, Classical Mechanics and Analog-Digital Electronics are necessary to build the foundation for advanced course. The students are expected to understand the fundamental concepts and develop ability to solve numerical problems. In addition, General Physics and Electronic practical are included to develop experimental hand. The course contents are elaborated as follows-

MPH -511: MATHEMATICAL METHODS IN PHYSICS (Marks: Max.-80, Min. – 32)**UNIT-I: Complex Variables**

Cauchy Riemann conditions, Analytic function, Line Integrals, Cauchy Integral Formula, Derivatives of Analytical functions and Fundamental theorem of Algebra. Taylor's series, Laurent's series, Integration and Differentiation of Power series, Zeroes of Analytical Functions, Singular Points-residues, Cauchy's residue theorem, poles.

UNIT-II: Differential Equations

Solutions of Differential Equation in series, Legendre, Bessel, Hermite, Legendre functions and their generating functions, Recurrence relations and Orthogonal Properties, Confluent Hyper-geometric functions, Laplace, Poisson and wave equations and their solutions with boundary conditions.

UNIT-III: Integral Transforms

Fourier Transform, Transforms of Derivatives, convolution Theorem, Parseval's Theorem, Laplace Transform of simple function and derivatives, Properties of Substitutions, determination of Inverse Transform, Laplace Transforms solution of differential equations.

UNIT-IV: Computational Methods in Physics

Linear systems of Equation, Matrices, Matrix Inversion, Eigen Value problems, Interpolation, Numerical Differentiation and Integration, Numerical Solution of Ordinary Differentiation equations, Runge-Kutta and Matrix Methods

UNIT-V: Tensors

Tensors- Notations and Conversions, Contravariant tensors, Rank of the Tensors, Properties of the Tensors e.g. Addition, Subtraction and Product, Contraction, Cartesian tensors and their transformation properties, Eigen values of second rank tensors, Quotient law, Higher Rank Tensors with examples from piezoelectricity, stiffness and compliance.

BOOKS RECOMMENDED

1. Theoretical Physics	P. M. Morse & H. Feshbach
2. Mathematical Methods in Physics	George Arfken
3. Schaum Series of Complex variables, Group theory	Murray R. Spigel
4. Matrix and Tensors	A. W. Joshi
5. Complex Variables	Churchill and Brown
6. Applied Mathematics for Engineers and Physicists	L. A. Pipes
7. Mathematical Physics	P. Chattopadhyay
9. Mathematical Physics	Ghatak, Goyal and Guha
10. Numerical Analysis	Rajaraman

MPH-512: QUANTUM MECHANICS (Marks: Max.-80, Min. – 32)**UNIT-I: Basic Concepts**

Uncertainty and complementarity, Schrodinger's wave equation. One dimensional square well potential, Barrier Problem, Linear Harmonic Oscillator, Spherically symmetric potential in three dimension, Hydrogen atom, Principle of indistinguishability of identical particles. Pauli's exclusion principle.

UNIT-II: Operators and Matrix Theory

Hilbert space, Bra and Ket notations, Equation of motion, Schrodinger, Heisenberg and Interaction Pictures, Creation, annihilation and number operators, Matrix theory and harmonic oscillators. Angular momentum,

Commutation relations, Eigen values and Eigen functions of J^2 and L_z , Pauli's Spin Matrices, Stern Gerlach Experiment, addition of angular momenta, Clebsch-Gordan coefficients.

UNIT-III: Approximation Methods

The WKB approximation, Classical limit, approximate solutions, asymptotic nature of the solutions, Solutions near a turning point, linear turning point, Connection at the turning point, asymptotic connection formulas, Application to Eigen value problems.

UNIT-IV: Stationary Perturbation Theory

Stationary Perturbation theory, Non-degenerate case, First order perturbation, second order perturbation of an oscillator, Degenerate case, Removal of degeneracy in second order, Zeeman Effect without electron spin, First order Stark effect in Hydrogen.

UNIT-V: Time Dependent Perturbation Theory

Time dependent Perturbation theory, Application to the theory of Scattering, Transition probability, Fermi's golden rule, Adiabatic and Sudden Approximations, Ionization of hydrogen atom

BOOKS RECOMMENDED

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|--------------------------------------|----------------------------------|
| 1. Quantum Mechanics | L. I. Schiff |
| 2. Quantum Mechanics | E. M. Lifshitz & L.E. Landau |
| 4. Quantum Mechanics | A. K. Ghatak & S. Loknathan |
| 5. Quantum Mechanics | David Griffith |
| 6. Text book of Quantum Mechanics | P. M. Mathews and K. Venkateshan |
| 7. Introduction to Quantum Mechanics | L. Pauling and E. B. Wilson |
| 8. Quantum Mechanics | B.K. Agarwal and HariPrakash |
| 9. Quantum Mechanics | V. K. Thankappan |

MPH -513 : CLASSICAL MECHANICS (Marks: Max.-80, Min. – 32)

UNIT-I: Lagrangian Mechanics

Mechanics of a single & system of particles, Rocket equation, generalized coordinates. Galileo's relativity principle. The Lagrangian for a free particle. The Lagrangian for a system of particles. The Lagrangian for a system of particle.

UNIT-II: Conservation and Variational Principle

Laws of conservation as derived from homogeneity and isotropy of space and homogeneity of mass and principle of mechanical, Similarity, virial theorem, variational Principle, Lagrangian and Hamiltonian formalism, constraint Holonomia and non-Holonomic, D'Alembert Principle

UNIT-III : Scattering and Collision

Reduced mass, Motion in a central field, Kepler's problem, Scattering in the central field, scattering cross section. Rutherford formula. Elastic and inelastic collision. Small oscillations and normal modes, forced oscillation. Normal coordinates. Frequency of molecular vibration. Damped oscillation. Parametric resonance.

UNIT-IV: Rigid Body Motion

Motion of a rigid body. Euler's angles. Inertia tensors. Angular momentum of a rigid body. Precision Euler's equations. Symmetric and asymmetric top. Non-inertial frames and pseudo-forces.

UNIT-V: Canonical Equation

Canonical Equation. Hamilton's equations. Canonical transformations. Poisson brackets and Canonical transformations. Hamilton Jacobi theory. Action angle variables, Maupertuis principles. Adiabatic invariants.

BOOKS RECOMMENDED

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|--|--------------------------------------|
| 1. Classical Mechanics | H. Goldstein (Addison & Wesley). |
| 2. Classical Mechanics | Landau & Lifshitz (Pergamon Press) |
| 3. Introduction to Classical Mechanics | Takwale & Puranik (Tata McGraw Hill) |
| 4. Classical Mechanics | Rana and Jog (Tata McGraw Hill) |

MPH- 514: ANALOG AND DIGITAL ELECTRONICS (Marks: Max.-80, Min. – 32)**UNIT-I: Differential and Operational Amplifier**

Differential amplifier- circuit configurations; dual input, balanced output differential amplifier; DC analysis; AC Analysis, inverting and non inverting inputs; CMRR; constant current bias level translator Block diagram of a typical Op-Amp analysis, Open loop configuration inverting and non inverting amplifiers, Op-amp with negative feedback

UNIT-II: Practical Operational Amplifier

Practical op-amp input offset voltage; input bias current, total output offset voltage, CMRR frequency response. DC and AC amplifier summing, scaling and averaging amplifiers, instrumentation amplifiers, integrator and differentiator.

UNIT-III: Op-amp Oscillator

Oscillator principles; Oscillator types; frequency, stability; response; the phase shift oscillator; Wein bridge oscillator; LC tunable oscillators; multi-vibrators- monostable and astable; comparators; square wave and triangular wave generators,

UNIT-IV: Logic Gates

The transistor as a switch, OR, AND & NOT gates; NOR and NAND gates; Boolean Algebra; D'Morgan's theorem; Exclusive OR gate, Decoder/De-multiplexer, Data selector/ multiplexer; Encoder.

UNIT -V: Flip flop and Counters

Flip-flops; A 1-bit memory; RS flip-flops, JK Flip-flops; JK Master slave flip-flops; T flip-flops; D- flip-flops; Shift registers; Synchronous and asynchronous counters; cascade counters.

BOOKS RECOMMENDED

1. Electronic devices and circuit theory
2. Op-Amps and Linear integrated circuits
3. Digital Principle and Application

Robert Boylested & Louis
Ramakant Gaikwad
A. P. Malvino & D. P. Leach

COURSE SPECIFIC OUTCOME (CSO) IInd SEMESTER

The course content of second semester are designed to illustrate few more core subjects like Electromagnetism, Condensed Matter Physics and Statistical Mechanics. In continuation with the first semester one paper contains topics of Advanced Quantum Mechanics. The practical courses are on General Physics and Electronics Instrumentation.

MPH- 521: ELECTROMAGNETISM (Marks: Max.-80, Min. – 32)**UNIT-I: Polarization and Dielectric media**

Polarization, atomic and molecular polarizability, bound charges, field inside a dielectric, susceptibility, dielectric constant, electric displacement, Clausius-Mossotti relation, models of molecular polarizability, Electrostatic energy in dielectric media.

UNIT-II: Boundary Problems

Boundary value problem, Uniqueness of the solution of Laplace's equation, Boundary conditions for DE, Method of images, Images in dielectrics, Green's function for Poisson's equation.

UNIT-III: Magnetostatics

Macroscopic equation, boundary conditions on B & H, Methods of solving boundary value problems in magnetostatics, Uniformly magnetized sphere, Magnetized sphere in an external field, Permanent magnets, Spherical shell of permeable materials in a uniform field, Magnetic shielding.

UNIT-IV: Electromagnetic Field

Faraday's Law, Self and mutual inductance, Maxwell's equations scalar and vector potential, Lorentz and Coulomb Gauges, Energy and Momentum of an Electromagnetic field, Poynting Theorem, Plane electromagnetic waves in a non-conducting medium, Linear and circular polarization, Stoke's parameters. Reflection and refraction of electromagnetic waves at a plane interface.

UNIT-V: Relativistic Mechanics

Relativistic Mechanics: Proper time velocity, energy, momentum, kinematics, dynamics, relativistic electrodynamics: magnetism, field transform, field tensor, relativistic potential.

BOOKS RECOMMENDED

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|--|-------------------------------|
| 1. Classical Electrodynamics | J. D. Jackson |
| 2. Electrodynamics of continuous Media | L. D. Landau & E. M. Lifshitz |
| 3. Introduction to Electrodynamics | D. J. Griffiths |
| 4. Electromagnetics | B. B. Laud |

MPH- 522: CONDENSED MATTER PHYSICS (Marks: Max.-80, Min. – 32)**UNIT-I: Crystallography**

Point and Space group, Crystal structures- Hexagonal closed packed, Diamond, cubic structure and Perovskite structures, Reciprocal lattice, Brillouin Zone, X- ray diffraction, Bragg's law, Laue's equation, Reciprocal lattice vector. Fourier analysis of the basis, Scattered wave amplitude, Structure and form factors.

UNIT-II: Mechanical Properties

Mechanics of deformed bodies for cubic crystals, stress strain tensors, Compliance and stiffness constants, elastic constants and energy density of cubic crystals, elastic waves in crystals and elastic isotropy.

UNIT-III: Lattice Dynamics

Normal modes of mono atomic and diatomic lattice vibrations, Dispersion relations, Phonon density of states, Quantum theories of specific heats, An-harmonic effect, Equation of state of solids, Thermal expansion, Gruneisen relation.

UNIT-IV: Electronic and Thermal Properties

Sommerfeld free electron model, Density of states, Application of electronic specific heat, Wiedemann Franz Law, Boltzmann Transport equation, Relaxation time approximation and application to electrical conductivity, Hall Effect.

UNIT–V: 3D lattice vibrations

Vibration of three dimension lattice, coupling parameter in variance relations, frequency dispersion curves and its experimental method of determination. Neutron scattering

BOOKS RECOMMENDED

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|--|---------------------------------|
| 1. Introduction to Solid State Physics | C. Kittel |
| 2. Introduction to Solids | R. A. Levy |
| 3. Principles of theory of Solids | J. M. Zeeman |
| 4. Solid State Physics | L. V. Azaroff |
| 5. Solid State Physics | N. W. Asheroff and N. D. Mermin |
| 6. Solid State Physics | A. J. Dekker |

MPH- 523 :STATISTICAL MECHANICS (Marks: Max.-80, Min. – 32)

UNIT–I:Basic Concepts

Concept of statistical distribution, phase space, density of states, Lioville's theorem, micro-canonical, canonical and grand canonical ensembles, thermodynamic relations, partition function for micro canonical ensemble and applications to ideal gas.

UNIT-II:M-B Distribution

Maxwell-Boltzman (M-B) distribution formula for velocity, equi-partition law of energy. Translational, Rotational and Vibrational partition functions, equation of state, entropy and specific heat of ideal gases.

UNIT–III:B-E Statistics

Grand canonical partition function, Bose – Einstein (B-E) statistics, strong and weak degeneracy. Application of B-E statistics to specific heats of solids, Bose Einstein condensation and phase transition. Theory of Liquid Helium. Fermi Dirac (F-D) statistics, application to free electron theory of metal and thermionic emissions.

UNIT–IV:Phase Transitions - I

Fluctuations in ensembles, energy and density fluctuations, random one dimensional walk and Brownian motion, electrical noise (Nyquist theory), Phase transition of the second kind, Ising model on one dimension, structural phase change, critical exponent, scaling hypothesis, theory of critical phenomena, mean field theory.

UNIT–V:Phase Transitions-II

Landau theory of phase transition, critical indices, scale transformation and dimensional analysis, Correlation of space-time dependent fluctuations, fluctuations and transport phenomena, Brownian motion, Langewin theory, fluctuation dissipation theorem. The Fokker-Plank equation

BOOKS RECOMMENDED

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|--|---------------------------------|
| 1. Fundamentals of Statistical and Thermal Physics | F. Reif |
| 2. Statistical Mechanics | K. Huang |
| 3. Statistical Mechanics | B. K. Agarwal |
| 4. Statistical Mechanics | R. Kubo |
| 5. Statistical Physics | L. D. Landau and E. M. Lifshitz |
| 6. Introduction to Statistical Mechanics | B. B. Laud |

MPH- 524: ADVANCED QUANTUM MECHANICS (Marks: Max.-80, Min. – 32)

UNIT–I: Many Electron Problems

The many electron atom, One electron approximation, Hartree method applied to ground state of Helium, The cohesive energy of metal, Single particle wave function, Hartree-Fock method, Coulomb correlation and Fermi Hole.

UNIT–II: Scattering

The general theory of scattering, General formula for approximation method and its application to derive

scattering amplification and cross section. Validity of Born approximation, Rutherford scattering formula for Born approximation. Partial wave analysis, Radiative transitions in atoms, dipole transitions, Selection rules.

UNIT-III: Identical Particles

Identical particles; Symmetric and anti-symmetric wave functions; Collision of identical particles; Spin angular momentum; spin functions for a many electron system.

UNIT-IV: Radiation

Einstein coefficient, Atom field interaction, Spontaneous Emission Rates, the selection rules, quantum theory of radiation, quantization of Radiation field, Spontaneous and stimulated Radiations.

UNIT-V: Electromagnetic Potentials

Klein –Gordon Equation, Dirac’s relativistic equation: Free particle equation, matrices of alpha and beta. Free particle solutions negative energy states, charge and current densities. Electromagnetic potentials, Dirac’s equation for a central field, The hydrogen atom.

BOOKS RECOMMENDED

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|---|-----------------------------|
| 1. Quantum Mechanics | L. I. Schiff |
| 2. Quantum Mechanics | L. D. Landau E. M. Lifshitz |
| 3. Quantum Mechanics | Sakurai |
| 4. Theory and applications of Quantum Mechanics | A. Yariv |
| 5. Quantum Mechanics | A. S. Davydov |
| 6. Wave Mechanics of Electrons in Metals | Stanley Raimis |

COURSE SPECIFIC OUTCOME (CSO) IIIrd SEMESTER

In the third semester the students are expected to enhance the understanding of advanced level subject. The course contents of this semester include Nuclear Physics, Atomic and Molecular Physics, Advanced Condensed Matter Physics. The students have choice to opt for Digital System and Microprocessor or Semiconductor Devices paper. The practical session will include programming on microprocessor and computer as well.

MPH- 531: NUCLEAR PHYSICS (Marks: Max.-80, Min. – 32)**UNIT–I: Binding Energy**

Binding energy, Semi-empirical mass formula and Nuclear Stability, Gamow's theory of α -decay, Neutrino hypothesis, Fermi theory of beta-decay, gamma-decay, Internal Conversion, Interaction of charged particles with matter, Pair Production.

UNIT–II: Nuclear Forces

Ground and excited state of Deuteron, p-p, n-n, p-n scattering at low energies, charge independence and spin dependence of Nuclear Forces, Exchange Forces, Properties of Nuclear Forces, Liquid Drop, Shell and collective models of Nucleus.

UNIT–III: Nuclear Reactions

Different types of reactions and conservation laws, Energies of nuclear reactions, Iso-spin, Reaction-cross section, Scattering and reaction cross section, Compound Nucleus reaction, Fission, Reactors, Fusion processes, Characteristics of fusion.

UNIT–IV: Detectors

Betatron, Phase and Path stability, Synchro-cyclotron, synchrotron, Scintillation and Solid State Detectors, Cerenkov Detectors.

UNIT–V: Elementary Particles

Elementary particles, Conservation laws, Baryon and Lepto numbers, Iso-spin, hypercharge, Strangeness, Parity, Quark Model, Charm, Beauty and Truth, Quark Dynamics.

BOOKS RECOMMENDED

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|---------------------------------------|---------------|
| 1. Nuclear Physics | I.Kaplan |
| 2. Nuclear Physics- An Introduction | W.E. Burcham |
| 3. Introductory Nuclear Physics | K. S. Krane |
| 4. Introduction to Nuclear Physics | H. A. Enge |
| 5. Physics of the Nucleus | M. A. Preston |
| 6. Nuclear Physics – An Introduction. | S. B. Patel |

MPH- 532: ATOMIC AND MOLECULAR PHYSICS (Marks: Max.-80, Min. – 32)**UNIT–I: Basic Topics**

Quantum states of one electron atoms; Atomic Orbital; Hydrogen and Helium spectrum; Spectra of alkali elements; Spin orbit interaction and fine structure in alkali spectra.

UNIT-II: Zeeman Effect

Normal and Anomalous Zeeman effect; Paschen-Beck effect; Stark effect; Electron spin Resonance, Nuclear Magnetic resonance, LS and JJ Coupling; Hyperfine structure (qualitative)

UNIT–III: Types of Molecules

Types of molecules; Diatomic linear symmetric top, asymmetric top and spherical top molecules; Rotational spectra of diatomic molecules as a rigid rotor; Energy levels and spectra of non-rigid rotor; intensity of rotational lines; Stark modulated microwave spectrometer (qualitative).

UNIT–IV: Diatomic Molecules

Vibrational energy of diatomic molecule; Diatomic molecule as a simple harmonic oscillator; Energy levels and spectrum; Morse potential energy curve; Molecules as vibrating rotator; Vibration spectrum of diatomic molecule.

UNIT–V: Raman Effect

Raman Effect-Quantum Theory, Molecular polarizability, pure rotational spectra of diatomic molecules,

Vibration-rotation spectra, Experimental setup and simple applications.

BOOKS RECOMMENDED

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|---|--------------------|
| 1. Introduction to Atomic Spectra | H. E. White |
| 2. Fundamentals of molecular spectroscopy | C. B. Banwell |
| 3. Spectroscopy Vol I. II & III | Walker & Straughen |
| 4. Introduction to Molecular Spectroscopy | G. M. Barrow |
| 5. Spectra of diatomic molecules | Herzberg |
| 6. Molecular spectroscopy | Jeanne L McHale |
| 7. Molecular Spectroscopy | J. M. Brown |
| 8. Spectra of Atoms and Molecules | P. F. Bemath |
| 9. Modern Spectroscopy | J. M. Holias |

MPH- 533: ADVANCED CONDENSED MATTER PHYSICS (Marks: Max.-80, Min. – 32)

UNIT-I: Superconductivity

Superconductivity characteristic features, Critical current, Persistent current and Meissner effect, Critical Magnetic fields, Magnetic Susceptibility, flux quantization, specific heat, Thermal conductivity, Isotope effect, Optical energy gap, Quasi particle tunneling and Josephson effects (d.c.&a.c.), Electron phonon interaction, cooper pairs, BCS theory, Type I and II superconductivity, Introduction to high temperature superconductivity.

UNIT-II: Magnetism

Langevin's Diamagnetic and paramagnetic equations, Quantum theory of Para Magnetism, Ferromagnetism, Curie point and the exchange integral, saturation magnetization and its temperature dependence, ferromagnetic domains and Bloch wall.

UNIT-III: Dielectrics and Ferroelectrics

Macroscopic electric field, local electric field, Dielectric constant and polarizability, classification of ferroelectrics, ferroelectric domains, Piezoelectricity.

UNIT-IV: Semiconductors I

Energy bands: Direct and indirect band gap semiconductors, Intrinsic carrier concentration, p-type, n-type semiconductors, Fermi level, Carrier transport at low field, electrical conductivity and Hall effect.

UNIT-V: Semiconductors II

Generation, recombination, carrier injection, Schokley-Read-Hall theory of Recombination, Thermionic emission, space charge effect, p-n junction.

BOOKS RECOMMENDED

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|--|--------------------------------|
| 1. Physics of Semiconductor Devices | S. M. Sze and K. K. Ng. |
| 2. Solid State Physics | C. Kittel |
| 3. Solid State Physics | N. W. Aschcroft & N. D. Mermin |
| 4. Solid State Physics | S. O. Pillai |
| 5. Introduction to Solid State Physics | R. A. Levy |

MPH- 534: DIGITAL SYSTEM AND MICROPROCESSOR (Marks: Max.-80, Min. – 32)

UNIT-I: D/A, A/D Converter

Signal processing elements, DAC weighted resistor network, R-2R ladder network, ADC- Simultaneous, Counter type, Successive Approximation, single and dual slope; ADC and DAC specifications.

UNIT-II: Microprocessor

Introduction to microprocessor systems, Architecture of 8085, Buses, Registers, Arithmetic logic unit, Trends in microprocessor developments.

UNIT-III: Microprocessor programming

Assembly and higher level languages, Micro and Macro instructions, Addressing schemes, Instruction set for 8085, Assembly language programming using data transfer, Arithmetic and logic instructions, Stack and

subroutine, Assemblers, Interpreters and Compilers, debugging.

UNIT-IV:Microprocessor Interfacing

Interfacing, Dip switches, Seven segment display to 8085, General purpose programmable peripheral IC. 8255, Interfacing ADC and ACS case studies, Wave form generation; Temperature controller, 8085 Interrupts, Simple examples using SIM and RIM instructions.

UNIT-V:Data Communication

Basic concepts transmission format, Error Checks, Data Communication over telephone line), Standards in serial I/O. Software controlled serial I/O. 8085 SID and SOD lines.

BOOKS RECOMMENDED

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|---|-------------------------------|
| 1. Digital Principles and Applications | A. P. Malvino and D. P. Leach |
| 2. Microprocessors | Ramesh Gaonkar |
| 3. Microprocessors and Interfacing | Douglas V. Hall |
| 4. Digital circuits and systems | Douglas V. Hall |
| 5. Microprocessor and Digital Systems | Douglas V. Hall |
| 6. Microprocessors and Microcontrollers | B. P. Singh |

MPH- 535: SEMICONDUCTOR DEVICES (Marks: Max.-80, Min. – 32)

Unit I: Semiconductors

Semiconductors: Valence bond model of semiconductor- intrinsic and extrinsic semiconductors, the energy band model; p-n junction, depletion region and capacitance; the diode equation, I-V characteristics, temperature dependence, electrical breakdown in p-n junctions.

Unit II: Diode and Transistors

Zener and avalanche breakdowns; IMPATT, TRAPATT, PIN diode, Bi-polar Junction Transistors: Transistor action, the Ebers-Moll equations, CB, CE, CC configurations and characteristics, high frequency performance of transistor, alpha and beta cut-off frequencies, microwave transistor, switching transistor

Unit III: Unipolar Devices

Unipolar Devices: Metal-semiconductor contacts, the Schottky effect, JFET and MESFET, device characteristics, MOSFET, basic characteristics, charge-coupled devices (CCD), Optoelectronic Devices: Photovoltaic effect, the p-n junction solar cell, I-V characteristics, photo detectors: photoconductor, photodiode, avalanche photodiode; LEDs: radiative and non-radiative transitions; semiconductor LASERS, population inversion.

Unit IV: Rectifiers and Converters

Rectifying Circuit: Single Phase half wave, Bi-phase half wave, Three phase half wave, single phase Bridge, Three Phase Bridge rectifiers, Converters: Naturally commutated converters, Converter operation, Voltage source and Current source inverter, Inverter performance, Cyclo converter.

Unit V: Machines and Motors

Application: Variable speed DC drives, AC Machine, Synchronous Machine, Brushless Machine, Stepper Motors, Switched reluctance motor, Linear Motors, Switched mode power supplies, Series-resonant power supplies, Uninterrupted Power Supplies (UPS), Welding power supplies.

BOOKS RECOMMENDED

- | | |
|---|------------------|
| Introduction to Semiconductor Materials and Devices | M. S. Tyagi |
| Physics of Semiconductor Devices | Shur |
| Physics of Semiconductor Devices | S. M. Sze, |
| Solid State Electronic Devices | Ben G. Streetman |
| Power Electronics | P. S. Bhimbhra |
| Power Electronics | Cyril W. Lander |
| Power Electronics | Mohd.Rashid |

COURSE SPECIFIC OUTCOME (CSO) IVth SEMESTER

In the fourth semester, students study two compulsory papers viz., Electronic Devices and Instrumentation; Materials Science. They can choose either Fiber Optics and Laser Physics or Microwave and Optics Communication paper. There will be extensive project work on the topic of contemporary Physics or societal interest. Two practical, seminar and assignment will be the other components of study.

MPH- 541: Electronics Devices and Instrumentation (Marks: Max.-80, Min. – 32)**Unit-I: Semiconductor Devices**

Semiconductor Diodes (p-n junction, Zener, Tunnel, Gunn), Transistors, Field Effect Devices, Device Structure, Device Characteristics, Frequency Dependence and Applications

Unit-II: Photonic Devices

Radiative and non-radiative transitions; optical absorptions; Principle, operation and applications of Optoelectronic Devices (Solar Cells, Photo-Detectors, Light Emitting Diode)

Unit-III: Measurement Properties

Data Interpretation and Analysis, Precision and Accuracy, Error Analysis, Propagation of Errors.

Unit-IV: Instrumentation Design

Linear and Nonlinear Curve Fitting, Chi-Square Test, Transducers, Properties. Signal Conditioning and Recovery, Noise Reduction, Shielding and Grounding.

Unit-V: Vacuum Techniques

Units of Pressure Management, Characteristics of Vacuum, Applications of Vacuum, Vacuum Systems, Vacuum Pumps, Vacuum Gauges, Pumping Speed for a Vacuum System.

BOOKS RECOMMENDED

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|--|-------------------------------|
| 1. Semiconductor Devices- Physics and Technology | S. M. Sze |
| 2. Introduction to Semiconductor Devices | M. S. Tyagi |
| 3. Measurement, Instrumentation and Experimental Design in Physics and Engineering | M. Sayer & A. Mansingh |
| 4. Optical Electronics | Ajay Ghatak and K. Tyagarajan |
| 5. Materials Science of Thin Films | Milton Ohring |

MPH- 542: FIBER OPTICS AND LASER PHYSICS (Marks: Max.-80, Min. – 32)**UNIT-I: Fiber Optics I**

Propagation of light in optical fiber, basic structure and optical path of an Optical Fiber, Acceptance Angle, Numerical Aperture, Modes of Propagation, Attenuation in Optical Fiber, Absorption losses, Bending Losses, Radiation Losses, Pulse Dispersion, Materials Dispersion.

UNIT-II: Fiber Optics II

Chemical Vapor Deposition and Double Crucible Methods for fiber optics fabrication; Light Emitting Diode (LED) as a source; Fiber –LED coupling; Bandwidth and Spectral Emission of LED.

UNIT-III: Laser Theory I

Spontaneous and Stimulated emission, Einstein Coefficients, idea of light amplification: Threshold condition for light amplification, Pumping scheme, rate equation, three and four level Lasers; Monochromaticity, Directionality, Brightness, Coherence.

UNIT-IV: Laser Theory II

Active medium, Line shape broadening, Optical Resonance, Longitudinal and transverse modes in laser cavity, Oscillation gain and power output, Q-Switching, Mode locking, pulse shortening.

UNIT-V: Types and Applications of Lasers

Principles of Ruby, Nd: YAG, He-Ne, CO₂, Semiconductor and dye Laser; Nonlinear Optics (Harmonic Generation), Holography and its applications.

BOOKS RECOMMENDED:

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|----------------------------------|---------------------------|
| 1. Laser Theory and Applications | A. K. Ghatak & Tyagarajan |
| 2. Introduction to Laser Physics | O'Shea |

- | | |
|----------------------------------|---------------------|
| 3. Introduction to Laser Physics | K. Shrimoda |
| 4. Laser and Nonlinear Optics | B. B. Laud |
| 5. Laser Fundamentals | William T. Silfvast |

MPH- 543: MICROWAVE AND OPTICS COMMUNICATION (Marks: Max.80, Min. 32)

UNIT I: Microwave Generators

Microwave Generators and wave guides: Failure of vacuum tubes at high frequency, Two cavity klystron, reflex klystron oscillator, magnetron oscillator, TWT amplifier.

UNIT II: Oscillators

Backward wave oscillator, GaAs oscillator; Propagation of EM waves through wave guide, TE, TM and TEM waves.

UNIT III: Microwave Components

Microwave components and Measurements: Microwave components: scattering matrix, attenuators, Tees, directional couplers, circulators, isolators, phase shifters, cavity resonators

Microwave measurements: Measurement of VSWR, phase shift, frequency, power, attenuation, dielectric constants of liquids and solids, Q of cavity.

UNIT IV: Fiber Optics Communication I

Principles of optical communication, single mode and multi mode fibers, step index, graded index, ray model, multi path dispersion, material dispersion, optical fiber as wave guide, fiber sources and detectors.

UNIT V: Fiber Optics Communication II

Manufacture and Measurements of fibers: Optical fiber cable, fiber joints, splices, couplers and connectors, measurement in optical fibers, attenuation measurement, dispersion measurement, refractive index profile measurement, transmission links, optical transmitters and receivers.

BOOKS RECOMMENDED:

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|---|---------------|
| 1. Microwave Engineering | David Pozar |
| 2. Introduction to Microwave Theory and Measurement | A. L. Lance |
| 3. Optical Fiber Communication | B. Keiser |
| 4. Optical Communication Systems | J. Gower |
| 5. Fiber Optic Communication | D. C. Agrawal |

MPH- 544: MATERIALS SCIENCE (Marks: Max.-80, Min. – 32)

UNIT-I: Phase Transition

Relative stability of phase and phase rule, Single component and binary phase diagrams, Micro structural changes during cooling, Lever rule. Order, Disorder Transition.

UNIT-II: Kinetics and Diffusion

Nucleation Kinetics, Growth and transformation Kinetics, Application in transformation in steel, solidification and crystal growth; Diffusion in solids, Fick's law, Solution to Fick's second law, Kirkendal effect

UNIT-III: Nanomaterials

Basic concepts and applications, Types of Carbon based nanomaterials, Fullerenes, Carbon nanotubes, Single wall and multi-wall carbon tubes, Synthesis of carbon nanomaterials, Electronic and mechanical properties of nano-materials.

UNIT-IV: X-Ray Diffraction

X-ray Diffraction process and Diffractometer, applications of XRD, Principle of powder diffraction method, Interpretation of XRD data, accurate determination of lattice parameters; least-square method (Rietveld Analysis)

UNIT-V: Microscopy Techniques

Microstructure of materials, Scanning and Transmission Electron Microscopy techniques, compositional analysis by energy dispersive spectroscopy; surface analysis by Scanning Tunneling and Atomic Force Microscopy

BOOKS RECOMMENDED:

1. An introduction to X-Ray Crystallography
2. X-Ray Crystallography
3. Crystallography for Solid State Physics
5. The Powder method
6. Crystal Structure Analysis
7. Materials Science and Engineering
8. Elements of Materials Science and Engineering
9. Characteristics of Materials

M. M. Woolfson

Azaroff

A. R. Verma, O. N. Shrivastava

Azaroff and Buerger

Buerger

V. Raghavan

L. H. Van Vlack

Ed. E. Lifshin